

Stage 3 Judges' Rubric - DECI Level 2

Arduino + Web Integration Project - Stage 3 Judge Evaluation

Team Name / #

Project Name

Judge Name

Date

How to Score Every Row

Enter ONE number from 1 to 5. Every row is worth 5 points; harder sections have more rows and therefore more weight.

1	Weak	Missing, incorrect, non-functional, or little/no meaningful evidence.
2	Developing	Partially demonstrated; important gaps remain or substantial support is needed.
3	Meets Expectations	Core requirement is present, understandable, and works at an acceptable/basic level.
4	Strong	Complete, accurate, well integrated, and supported by good evidence; only minor issues.
5	Excellent	Highly effective, reliable, purposeful, well-tested/polished, and clearly justified by the team.

Judge the final Stage 3 work, using the previous-phase requirements as the baseline. **Flow:** reward evidence, correct integration, and purposeful design - not complexity for its own sake. **Testing:** look for real trials/results, not only claims. **Hardware:** reward the best-fit use of components and reliable integration. **Teamwork & presentation:** look for shared ownership, clear communication, and the ability of multiple members to answer questions.

Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Problem, Research & Solution			Section Weight: 15 points
Problem, users & research	Defines a real problem and target users with relevant research findings/references.		
Solution fit & user interaction	Solution directly addresses the problem and clearly explains how users interact with it.		
Features & value	Main functions are purposeful, coherent, and create a clear advantage or value for the user.		
System, Circuit, Hardware & Kit			Section Weight: 25 points
System data flow	Full path from sensor/input -> Arduino processing -> communication/output -> web/software is clear and correct.		

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Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
System, Circuit, Hardware & Kit (continued)			Section Weight: 25 points
Purposeful kit selection	Car/Robot Arm Kit choice is justified by the components actually needed; unnecessary kit parts are not forced into the project.		
Component roles	Arduino, sensors, actuators, communication modules, and other parts each have a clear and correct role.		
Circuit correctness	Power, ground, signal, TX/RX, and other relevant connections are technically reasonable and match the prototype.		
Best hardware use & reliability	Hardware is used effectively for the problem and performs consistently in repeated trials/demos.		
Programming & Control Logic			Section Weight: 15 points
Inputs, conditions & outputs	Sensor/input readings drive correct decisions and outputs.		
Loops & functions	Repeated behavior and reusable functions/blocks are used appropriately and clearly.		
Control reliability	Program behavior is stable, understandable, and consistent with the flowchart/pseudocode.		
Web Interface Integration			Section Weight: 15 points
Web role & system connection	Web interface displays/controls the exact information produced by the system and has a clear purpose.		

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Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Web Interface Integration (continued)			Section Weight: 15 points
UI quality & usability	Interface is readable, organized, and appropriate for the target users.		
Working integration evidence	Screenshots/demo show that the web interface and hardware/data flow are credibly connected or implemented.		
Testing & Hardware Evidence			Section Weight: 10 points
Evidence of testing	Shows repeated hardware/software tests, measurements, logs, screenshots, or scenario results.		
Improvement & robustness	Team uses findings to improve thresholds, communication, UI, wiring, reliability, or error handling.		
Business & Value			Section Weight: 5 points
Customer, alternatives & value	Identifies customer, existing alternatives/competitors, and explains practical value/differentiation.		
Teamwork & Presentation			Section Weight: 15 points
Shared contribution & roles	Team members visibly contribute across hardware, code, web, testing, or documentation.		
Technical communication	Explains system flow and design decisions clearly without relying on one member for all technical answers.		

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Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Teamwork & Presentation (continued)			Section Weight: 15 points
Presentation, demo & Q&A;	Presentation is clear, well paced, evidence-based, and questions are answered confidently.		

Extra Requirements / Optional Bonus			Optional Bonus - max 10
AI / data feature	Optional AI, ML, or data-analysis feature uses relevant project data and includes a useful result or visualization.		
Advanced system feature	Optional cybersecurity, automation, IoT, or software enhancement strengthens the existing system and includes evidence.		

Score Summary

Section	Maximum	Section Subtotal	Notes
Problem, Research & Solution	15		
System, Circuit, Hardware & Kit	25		
Programming & Control Logic	15		
Web Interface Integration	15		
Testing & Hardware Evidence	10		
Business & Value	5		
Teamwork & Presentation	15		
CORE TOTAL	100		

OPTIONAL BONUS (record separately, max 10)	
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